

Measuring Mitosis

Teacher guide | Biology 2e, Chapter 10 | 40-55 minutes

Learning targets and timing

What can a snapshot of dividing cells tell us about the cell cycle?

Classify cells in two schematic root-tip fields or count prepared classroom slides. Use stage frequencies to estimate time in each stage under an explicitly assumed cycle duration.

Suggested sequence: 8-10 min prediction and setup; 20-30 min observation; 12-20 min analysis. Virtual time can be accelerated. For photosynthesis and yeast, pool independent replicates from class groups; three sequential repeats need extra class time.

Route choices and sharing

In Teacher setup, select Virtual or Classroom for each treatment/field and create an assignment link. Assign all virtual for absences or a substitute. For a hybrid route, conduct one part in class and simulate the others; compare data-source limitations before pooling. These links are editable and do not restrict access.

Materials

- Prepared onion root-tip slides, microscope, stage tally sheet
- Alternatively: the app's two schematic fields, with no equipment needed
- Calculator; teacher-selected cycle duration (24 hours is an assumption, not a measured value)

Classroom procedure

1. Review the reference diagrams and stage definitions. Interphase includes G1, S, and G2; these are not distinguished in this activity.
2. Locate a root-tip meristem on a prepared slide at low magnification, then increase magnification to resolve chromosomes. Use fine focus as appropriate.
3. Choose a systematic field and count every interpretable cell without selecting only dividing cells. Record excluded or ambiguous cells in notes.
4. Classify cells as interphase, prophase (including prometaphase), metaphase, anaphase, or telophase. Record a second non-overlapping field.
5. Calculate each stage count / total count x 100. Multiply that fraction by the assumed cycle duration. Explain the assumptions needed to turn a snapshot into a time estimate.

Safety: Use prepared slides only. Carry microscopes with two hands and handle slide edges carefully. The classroom route does not require stains, acids, or cutting root tips.

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Expected reasoning and troubleshooting

The schematic fields together contain 27 interphase, 6 prophase, 3 metaphase, 2 anaphase, and 2 telophase cells (40 total). Order is shuffled per attempt. Review feedback appears only after all cells in a virtual field are classified.

At an assumed 24-hour cycle, the modeled durations are 16.2, 3.6, 1.8, 1.2, and 1.2 hours respectively. These are worked examples for the schematic sample, not reference durations for real roots.

Score the quality of morphological evidence and limitations. Classroom counts should not be corrected to match the schematic distribution. In this simplified key, prometaphase is pooled with prophase.

Model boundaries

The virtual fields are original teaching diagrams, not microscope photographs or empirical samples. Each contains 20 cells. They practice recognition and frequency calculations; the curated distribution is not a validated biological estimate.

Estimated stage time = stage count / total count x assumed cycle duration. This assumes representative sampling of an asynchronous, steadily cycling population with comparable cycle lengths. Non-cycling cells, sampling bias, and differential cell loss can invalidate the estimate.

Suggested assessment - 12 points

Dimension	Evidence to look for	Points
Design	Reasoned prediction; variables, controls or sampling plan	0-3
Data	Complete observations, correct units, source labels, repeats	0-3
Analysis	Appropriate calculations and comparisons supported by data	0-3
Explanation	Biological reasoning, uncertainty, and a feasible improvement	0-3

For each dimension: 0 = absent; 1 = partial or major errors; 2 = mostly sound with a minor omission; 3 = complete and well supported. Suggested teacher rubric only; the app does not assign a grade.

Downloads, access, and classroom pilot

Student worksheet: three printable pages. Teacher guide: procedures, interpretation, and rubric. Reports and CSV data download from Analyze & explain. Teacher guides are public resources, not protected answer keys. Before classroom use, pilot the equipment, timing, and reading level. Editable Word and fillable PDF versions are planned.

Reading and method connections

OpenStax Biology 2e, 10.2: Open source

Text and diagrams in these labs are original teaching materials. Simulated observations are not measurements from the linked sources.